

# Tanmay Thakur

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## SKILLS

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**Languages** : Python, C++, JavaScript, TypeScript, SQL

**Systems & Architecture** : Distributed Systems, Asynchronous Processing, Concurrency, Multi-threading, Synchronization, Software Design, System Design, API Design, Celery

**Frameworks & Libraries** : FastAPI, React.js, Tailwind CSS, Node.js, Express.js

**AI & Machine Learning** : TensorFlow, Keras, scikit-learn, Deep Learning, Retrieval-Augmented Generation (RAG), Time Series Forecasting, Vector Embeddings

**Databases** : PostgreSQL, MongoDB, Redis

**Developer Tools** : Git, GitHub, Docker, Linux, Figma

**Computer Science Fundamentals** : Data Structures & Algorithms, Object-Oriented Programming (OOP), Operating Systems, Database Management Systems (DBMS), Computer Networks (CN)

## WORK EXPERIENCE

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### Indian Society of Technical Education - NITH

September 2024 – August 2025

*Member*

NIT, Hamirpur

- Collaborated with student teams across technical and organizational domains, contributing to society initiatives and technical event planning.
- Assisted in organizing and hosting events for more than 1000+ participants in the annual technical festival, supporting logistics, coordination, and participant engagement.
- Worked with cross-functional teams to ensure smooth execution of society activities and student-led technical programs.

### Student Council, Ryan International School, Ghaziabad

May 2022 – March 2023

*President*

Ryan International School, Ghaziabad

- Led a 40-member student council representing over 2000 students, coordinating school-wide events and administrative initiatives.
- Organized and supervised multiple school events, ceremonies, and student engagement programs.

## PROJECTS

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### LearnTube 🎧 – FastAPI, PostgreSQL, Redis, Celery, Retrieval-Augmented Generation (RAG), Vector Embeddings

- Built a distributed AI-powered learning platform using FastAPI, PostgreSQL, Redis, and Celery, leveraging asynchronous task queues and semantic retrieval to enable conversational interaction with YouTube videos.
- Designed an asynchronous transcript processing pipeline using Celery workers, semantic chunking, and vector embeddings to efficiently process long-form videos and generate context-aware responses with timestamped citations.
- Developed a real-time AI chat interface with semantic search and streaming responses, reducing information retrieval from hour-long educational videos to seconds.

### Scalora 🎧 – React, FastAPI, TensorFlow, Scikit-Learn, LSTM, Isolation Forest

- Built Scalora, an AI-powered cloud cost optimization platform using LSTM and Isolation Forest to forecast infrastructure demand and simulate intelligent scaling decisions.
- Designed and implemented a full-stack cloud optimization system using React, FastAPI, and TensorFlow to predict CPU utilization and estimate real-time infrastructure cost savings.
- Implemented dynamic cost modeling based on provider, instance type, and scaling duration to quantify financial impact.

### FanBase 🎧 – React, TypeScript, Node.js, Express, MongoDB, Tailwind CSS

- Built a full-stack content monetization platform where creators can upload exclusive posts and users can subscribe to access premium content using secure authentication and role-based access control.
- Designed and implemented secure REST APIs with JWT authentication, media uploads, and MongoDB integration for scalable content management.
- Built a responsive React frontend with role-based access control, optimized media rendering, and seamless content browsing for creators and subscribers.

## EDUCATION

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### National Institute of Technology, Hamirpur

*Bachelor of Technology*

Aug 2024 – Present

Hamirpur, Himachal Pradesh